Current Date: March 29, 2026

WORKSHOP PLAN

Facilitation Guide: Introduction to Machine Learning

Summit: Ghana Data Science Summit (IndabaX Ghana) 2026

Themes: *Data First: Unlocking Ethical and Inclusive AI | From Skills to Impact*

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1. Session Objectives & Target Audience

The primary goal of this workshop is to demystify Machine Learning (ML) for Ghanaian professionals and students, moving beyond theory to demonstrate how data-driven decisions create tangible impact in the local economy.

Key Objectives:

- **Conceptual Mastery:** Understand the ML lifecycle from data collection to model deployment within an ethical framework.
- **Contextual Application:** Identify specific use cases in Ghana's agriculture, fintech, and healthcare sectors.
- **Ethical Awareness:** Recognize and mitigate bias in datasets to ensure inclusive AI solutions.
- **Skill-to-Impact Bridge:** Learn how to transition from building models in a sandbox to solving real-world community problems.

Target Audience:

- Final-year undergraduate and graduate students in STEM.
- Early-career software engineers looking to pivot into Data Science.
- Policy makers and industry leads seeking to understand the technical foundations of AI impact.

2. Part 1: Foundations & Intuition (0:00 - 1:30)

This segment focuses on building the "ML Mindset" and establishing the critical importance of data quality and ethics before touching code.

0:00 - 0:20 | Introduction & The "Human ML" Icebreaker

- **Activity:** *The Sorting Game.* Ask five volunteers to come forward. The audience must "train" an algorithm (one person) to sort these volunteers by a specific metric (e.g., height, or tenure in tech) using only "Yes/No" questions.
- **Talking Point:** Define ML as "teaching computers to learn from patterns rather than following explicit instructions."

0:20 - 0:50 | The ML Taxonomy through a Ghanaian Lens

Type	Concept	Local Case Study
Supervised	Learning from labeled data.	Predicting **Cocoa Pod** ripeness based on image color data.
Unsupervised	Finding hidden patterns.	Segmenting **Mobile Money (MoMo)** users into spending personas for targeted financial literacy.
Reinforcement	Learning via trial and error.	Optimizing **Traffic Signal** timing at the Kwame Nkrumah Interchange in Accra.

0:50 - 1:30 | "Data First": Ethics, Bias, and Inclusion

> **Case Study Discussion:** "The Credit Scoring Dilemma."
>
> Imagine a bank uses an ML model to grant loans. If the historical data only includes urban males from Accra, what happens to a female shea-nut farmer in the Northern Region?
>
> **Facilitator Note:** Use this to introduce *Inclusive AI*. Discuss how "Data First" means ensuring data representativeness across Ghana's diverse linguistic and regional demographics.

- **Key Concept:** Garbage In, Garbage Out (GIGO).

- **Inclusion Focus:** The importance of NLP for local languages (Twi, Ga, Ewe) to make AI accessible to the non-English speaking population.

🕑 **SCHEDULED BREAK: 1 Hour 30 Minutes Mark (15-20 Minutes)**

Encourage networking and "Problem Wall" submissions.

3. Part 2: Practical Application & Local Impact (1:50 - 3:00)

The second half shifts from "What is it?" to "How do we make it work for Ghana?"

1:50 - 2:20 | The ML Lifecycle: From Skills to Impact

Walk through the technical workflow using a specific project: **Predictive Maintenance for GRIDCo Power Grids.**

1. **Problem Definition:** Reducing "Dumsor" (power outages) through proactive sensor data.
2. **Data Acquisition:** IoT sensors on transformers.
3. **Feature Engineering:** Incorporating Ghanaian weather patterns (Harmattan dust levels, humidity).
4. **Evaluation:** Why *Precision* matters more than *Accuracy* in power grids.

2:20 - 2:45 | Interactive Group Challenge: "Impact Design"

Divide participants into small groups. Assign each group a sector: *Healthcare, Agriculture, Logistics, or Education.*

- **Task:** Design a high-level ML solution for a specific Ghanaian problem.
- **Constraint:** They must identify one potential **Ethical Risk** (e.g., privacy of patient data) and one **Inclusion Strategy** (e.g., voice-interface for illiterate users).

2:45 - 3:00 | Roadmap: The Journey Ahead

- **From Learner to Contributor:** Encouraging participation in open-source projects like *Sauti Yetu* or local Kaggle competitions.
- **Closing Charge:** ML is not a magic wand; it is a tool. The impact depends on the user's understanding of the local context.

4. Delivery Tips & Interactive Elements

- **Visual Aids:** Use minimal text on slides. Use photos of Ghanaian markets, farms, and cityscapes to ground the abstract concepts in reality.
- **The "Jargon-Free" Rule:** For every technical term (e.g., *Overfitting*), provide a real-world analogy (e.g., "A student who memorizes past questions but fails when the exam questions change slightly").
- **Live Polling:** Use Mentimeter or Slido to ask: *"Which sector in Ghana will be most transformed by AI by 2030?"* to keep engagement high.
- **Inclusive Language:** Ensure examples represent both the tech-savvy urban centers and the essential rural industries.

5. Resources & Next Steps for the Community

To move "From Skills to Impact," participants need a continuous learning path post-summit.

Recommended Learning Path:

- **Foundations:** *Mathematics for Machine Learning* (Linear Algebra and Calculus).
- **Tools:** Python (Scikit-Learn, Pandas) and the emerging 2026 low-code AI tools for rapid prototyping.
- **Community:** Join the **IndabaX Ghana** Slack/WhatsApp groups and attend monthly **AI Ghana** meetups.

Local Data Sources:

- **Ghana Open Data Initiative (GODI):** For government-related datasets.
- **Kaggle Africa:** For continent-specific datasets and challenges.
- **Zindi Africa:** The premier platform for African data science competitions.

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Reference

[1] Under the theme “Data First: Unlocking Ethical and ...
https://www.instagram.com/p/DV841bBjCqi/

[2] The Ghana Data Science Summit - IndabaX Ghana
https://blog.discoursechannel.com/events/the-ghana-data-science-summit-indabax-ghana/